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Project Title Generating electricity by harnessing the energy produced from walking.

Course Name and Code Linear integrated circuits

Course code :19EECC203

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Abstract: A brief summary of the project, including objectives, methodology, results, and relevance to SDGs.

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Introduction: The objective of the visit was to understand the application of advanced sensor technologies, particularly piezoelectric sensors, in real-world scenarios. The focus was on how these sensors can be used in commercial environments, such as shopping malls, to count the number of visitors through their footsteps, thereby contributing to sustainable development.

Introduction to the Company

- **1.1 Background**

- Introduction to the course project topic.
- This report focuses on the implementation of piezoelectric sensors for counting the number of people entering a shopping mall through a visitor counter system. The objective is to explore how piezoelectric sensors can be utilized to enhance visitor counting accuracy, contributing to the mall's operational efficiency and sustainable development. The project aims to design and analyze a system that leverages the piezoelectric effect to detect footsteps and provide real-time data on visitor numbers.

Working Principle

1. **Installation:** Piezoelectric sensors are embedded in the floor at strategic locations like entry and exit points to ensure that each footstep is detected.
2. **Detection:** When a person steps on the sensor, the pressure from the footstep generates an electrical charge in the piezoelectric material.
3. **Signal Processing:** The generated electrical signals are transmitted to a data acquisition system equipped with linear ICs. These ICs amplify and filter the signals to improve their quality and remove noise.
4. **Data Analysis:** The processed signals are analyzed using algorithms that differentiate between human footsteps and other pressures, such as trolleys or luggage, to ensure an accurate count.
5. **Display and Monitoring:** The number of visitors is displayed on digital counters and stored in a database for further analysis. This data is accessible in real-time for monitoring and decision-making.

Importance of linear integrated circuits in modern technology.

Linear integrated circuits (ICs) are essential components in modern electronics. They play a crucial role in processing analog signals, which are continuous signals like sound, light, or radio waves. Here's why they are important:

1. **Signal Amplification:** Linear ICs can amplify weak signals to make them stronger and more usable. This is vital in devices like hearing aids, radios, and televisions.
2. **Signal Processing:** They help in processing and conditioning signals, making them suitable for further use. For example, they can filter out noise from a signal to ensure clarity and accuracy.
3. **Versatility:** Linear ICs are used in a wide range of applications, from audio equipment and medical devices to communication systems and industrial machinery.
4. **Efficiency:** They make electronic systems more efficient by reducing the number of separate components needed. This leads to smaller, more reliable, and more cost-effective devices.
5. **Performance:** By integrating multiple functions into a single chip, linear ICs improve the performance and reliability of electronic circuits.

• 1.2 Objectives

- Clear statement of the project objectives.

The main goal of this project is to create a system that uses piezoelectric sensors to count the number of people entering a mall. Here are the specific objectives in simple words:

- **Count People Accurately:** Use sensors to reliably count how many people walk into the mall.
- **Real-Time Information:** Provide up-to-the-minute data on the number of visitors.
- **Save Energy:** Make the system energy-efficient by using sensors that generate power from footsteps.
- **Support Sustainability:** Help the mall be more eco-friendly by using durable and non-toxic sensor materials.
- **Work with Current Systems:** Ensure the new counting system works well with the mall's existing management systems.
- **Process Data Effectively:** Use technology to ensure the footstep signals are clear and accurate.
- **Easy to Use:** Create a simple interface for mall staff to see and understand the visitor counts
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- **1.3 Relevance to SDGs**
 - Explanation of which Sustainable Development Goals the project addresses and how.

Affordable and Clean Energy (SDG 7):

- The piezoelectric sensors used in this project convert mechanical energy from footsteps into electrical signals without the need for an external power source. This not only reduces energy consumption but also demonstrates the potential for renewable energy technologies in everyday applications.

Industry, Innovation, and Infrastructure (SDG 9):

- Implementing this advanced sensor technology in shopping malls demonstrates a commitment to innovative industrial practices. It enhances the mall's infrastructure by providing precise and real-time data on visitor traffic, which can be used to optimize various operational aspects such as staffing and energy management.

Sustainable Cities and Communities (SDG 11):

- By providing accurate data on foot traffic, the project helps mall management optimize resource allocation, improve safety, and enhance the overall visitor experience. Efficiently managed malls contribute to the development of sustainable and resilient urban environments.

Responsible Consumption and Production (SDG 12):

- The use of durable, non-toxic materials in the piezoelectric sensors ensures long-term use and minimal environmental impact. This aligns with sustainable production practices and encourages the responsible consumption of resources.

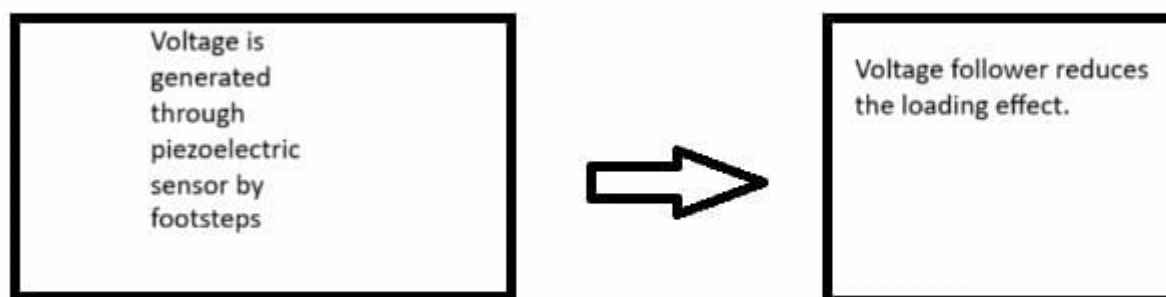
Climate action (SDG 13):

- The energy-efficient design of the visitor counting system reduces the mall's energy consumption and its carbon footprint. By integrating sustainable technologies, the project supports broader climate action initiatives aimed at mitigating the effects of climate change.

1. Methodology

1.1 Project Design:

1.1.1 Block Diagram representation of the project



1.1.2 Explanation of the block diagram

Voltage is generated through piezoelectric sensor by footsteps:

- When a person steps on the piezoelectric sensor, mechanical stress is applied to the sensor.
- The piezoelectric sensor converts this mechanical stress into an electrical voltage.
- This generated voltage serves as a signal indicating that a person has entered the mall.

Voltage follower reduces the loading effect:

- The generated voltage from the piezoelectric sensor might be weak and susceptible to loading effects, which can distort the signal.
- A voltage follower, which is typically an operational amplifier (op-amp) configured as a buffer, is used to mitigate this issue.
- The voltage follower provides high input impedance and low output impedance, ensuring that the signal from the piezoelectric sensor is preserved without being affected by subsequent stages in the circuit.

By the produced voltage, number of entries is counted and displayed on seven-segment:

- The voltage signal, now stabilized by the voltage follower, is fed into a counting circuit.
- The counting circuit registers each valid pulse (voltage signal) as a count, incrementing the visitor counter.
- The current count of visitors is then displayed on a seven-segment display.
- The seven-segment display visually represents the number of people who have entered the mall, making it easy to read the count at a glance.

Components used

The **piezoelectric sensor** generates a voltage when a footstep is detected.

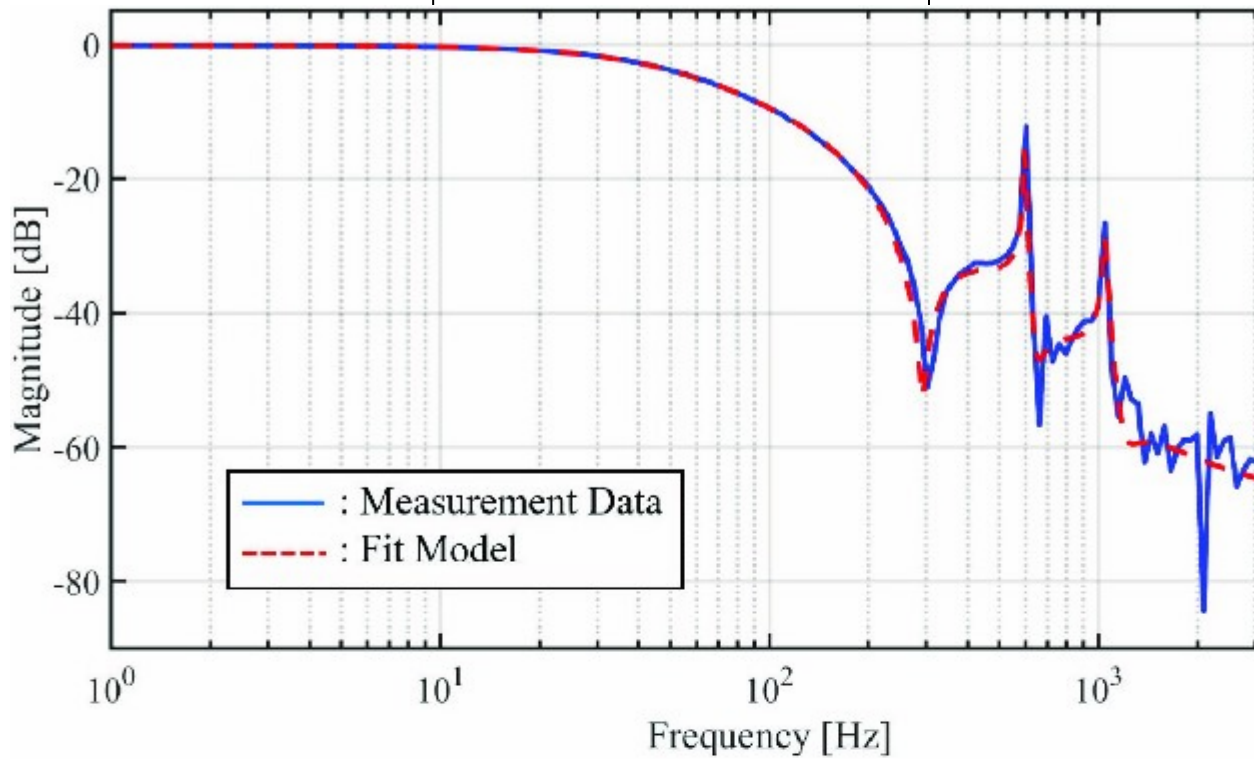
This voltage is fed to a **voltage follower** (op-amp) to ensure the signal integrity is maintained by reducing the loading effect.

A **photodiode** could be used to provide an additional detection mechanism, possibly verifying the presence of an individual before counting the entry.

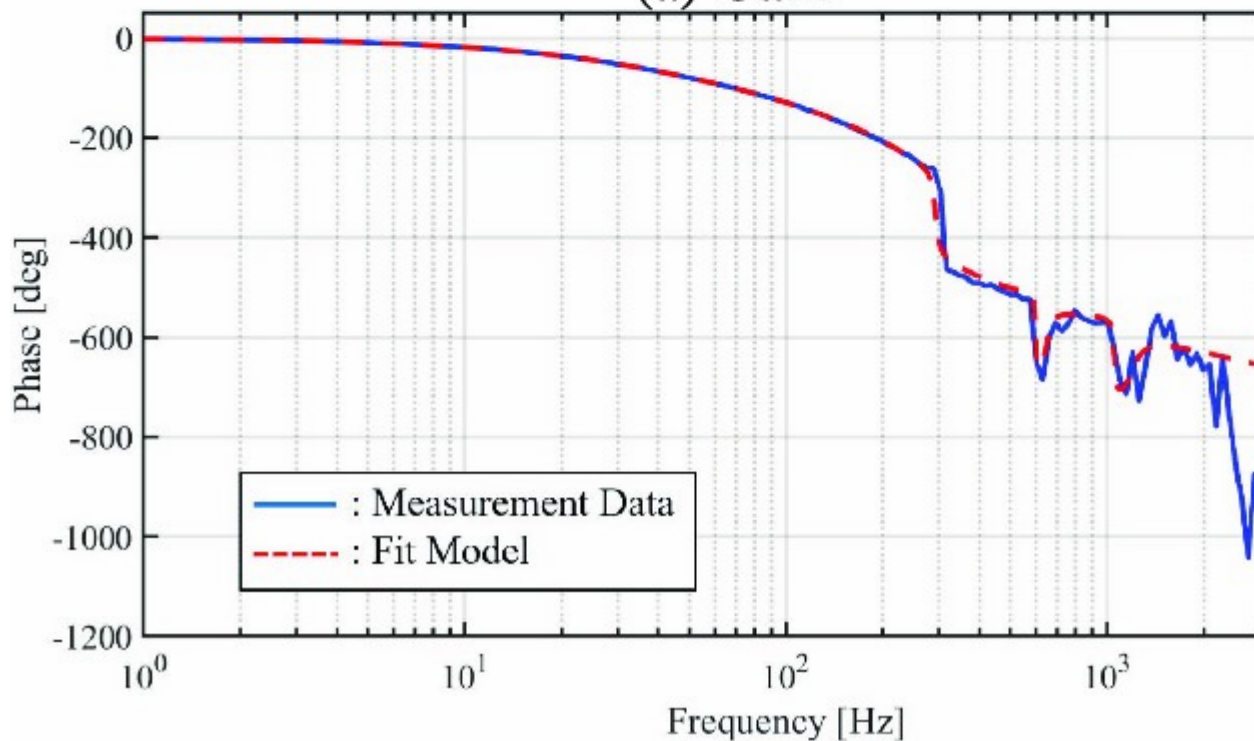
The combined signal is then sent to a **digital counter**.

The **counter** keeps track of the number of entries, which is displayed on a **seven-segment display**.

Bode plot of piezoelectric sensor:



(a) Gain



Piezoelectric sensor:

Force (N)	Voltage (V)
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0	0
1	0.5
2	1

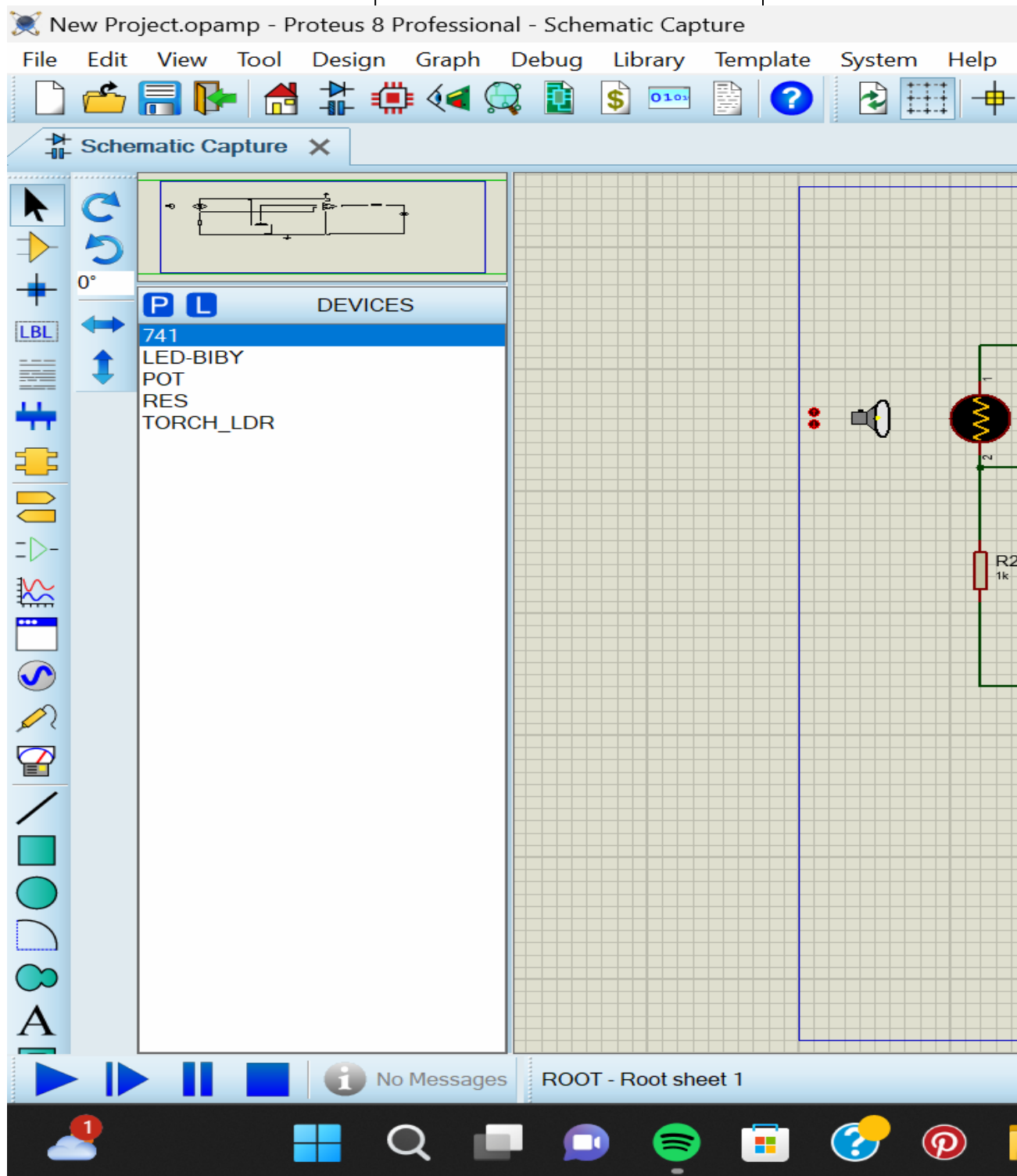
1. Circuit Design and Simulation

Circuit Diagram: Explanation and design of the circuit and listing the specifications of the circuit components

1.1 Simulation setup and tools.

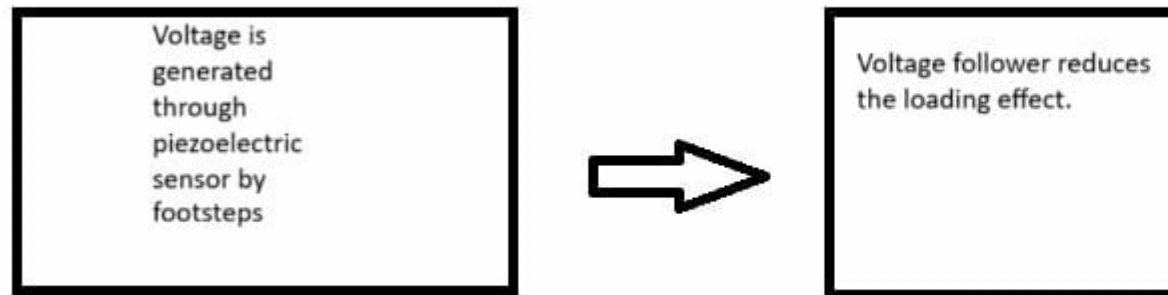
1.1.1 Detailed schematic of the Op-Amp circuit.

1.1.2 Description of the simulation environment (ex: Spice software, Proteus, Eagle)



Implementation

1.1.3 Flow chart of the proposed statement



1.1.4 Step-by-step process of the project implementation.

The piezoelectric sensor generates a small AC voltage when mechanical pressure, such as a footstep, is applied.

This voltage needs to be rectified and possibly amplified for further use.

A lot of people walk in and out of a mall. And we know that footsteps produce kinetic energy. We can use this kinetic energy to produce voltage sufficient to power people counter machine. The project does the same. It converts the kinetic energy produced by the footsteps to voltage and that voltage is used to power the people counter machine.

1.1.5 Snap shots of the circuit Bread Board and PCB



2. Results and Discussions

2.1 Data and observations from practical implementation.

We observed that we can use the voltage generated by footsteps to power the people counter machine.

2.2 Comparison with simulation results.

3. Overview of SDG

3.1 Explanation of SDG in general

SDG stands for Sustainable Development Goals. These are like a global to-do list for making the world a better place by 2030. There are 17 goals in total, covering things like ending poverty, ensuring good health and education for everyone, promoting gender equality, protecting the environment, and creating peaceful and just societies. Each goal has specific targets to achieve, and

they're all interconnected, meaning progress in one area can help improve others. The idea is to create a more fair, healthy, and sustainable world for everyone.

3.2 Discussion on how the project outcomes contribute to the targeted SDGs (SDG 7).

The project outcomes can significantly contribute to achieving Sustainable Development Goal 7 (SDG 7), which aims to ensure access to affordable, reliable, sustainable, and modern energy for all. By counting the number of people using a visitor counter powered by piezoelectricity, several SDG 7 targets can be addressed:

4. **Access to Energy:** Piezoelectricity, the conversion of mechanical energy into electrical energy, provides a sustainable source of power for the visitor counter. This technology can be deployed even in remote areas where access to conventional energy sources might be limited.
5. **Energy Efficiency:** Piezoelectric materials have the advantage of being highly efficient in converting mechanical stress into electrical energy. By harnessing this efficiency, the project promotes the use of energy-efficient technologies, contributing to SDG 7's target of doubling the global rate of improvement in energy efficiency by 2030.
6. **Renewable Energy:** Piezoelectricity is a form of renewable energy that can be harvested from various sources such as footsteps, vibrations, and even ambient mechanical movements. By utilizing piezoelectric materials to power the visitor counter, the project aligns with SDG 7's objective of increasing the share of renewable energy in the global energy mix.
7. **Monitoring and Accountability:** The visitor counter, powered by piezoelectricity, provides a means to monitor foot traffic in various locations. This data can be crucial for urban planners, businesses, and policymakers to optimize resource allocation, improve infrastructure planning, and enhance overall energy management strategies, thus contributing to SDG 7's target of enhancing international cooperation to facilitate access to clean energy research and technology.
8. **Innovation and Technology Development:** By employing piezoelectric technology in a practical application like a visitor counter, the project fosters innovation and technological development in the field of sustainable energy. This aligns with SDG 7's target of enhancing international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency, and advanced and cleaner fossil-fuel technology.

8.1 Any limitations and potential improvements

1. **Sensitivity to Movement:** Piezoelectric sensors can sometimes be too sensitive, leading to inaccuracies in counting when there are slight vibrations or movements nearby. This means the counter might count people who aren't actually passing by.
2. **Weather Dependence:** Piezoelectricity relies on mechanical pressure, so it might not work as well in extreme weather conditions like heavy rain, snow, or strong winds. This could affect the accuracy of the counting during these times.

3. **Maintenance Needs:** Piezoelectric sensors require periodic maintenance to ensure they're functioning correctly. Dust, dirt, or damage to the sensors can affect their performance over time, leading to inaccurate counting.
4. **Limited Range:** The range of piezoelectric sensors might be limited, meaning they can only detect people within a certain distance. If the area to be monitored is large, multiple sensors might be needed, increasing the complexity and cost of the system

Potential improvements could include:

1. **Enhanced Sensor Design:** Developing more robust and accurate piezoelectric sensors that are less prone to false readings and can withstand various environmental conditions better.
2. **Weatherproofing:** Creating weatherproof enclosures or protective covers for the sensors to shield them from harsh weather elements, ensuring consistent performance regardless of the weather.
3. **Regular Maintenance Schedule:** Implementing a regular maintenance schedule to clean and inspect the sensors, ensuring they remain in optimal working condition and replacing any damaged components promptly.
9. **Conclusions:** Write the summary of the project, remarks on contribution to SDGs.

The project aims to count the number of people using a visitor counter powered by piezoelectricity. Piezoelectricity is a technology that converts mechanical pressure into electrical energy. The visitor counter utilizes sensors that generate electricity when people walk over them, allowing the system to detect and count the number of visitors passing through an area.

Contribution to SDGs: This project contributes to several Sustainable Development Goals (SDGs):

1. **SDG 7: Affordable and Clean Energy:** By harnessing piezoelectricity, the project promotes the use of clean and sustainable energy sources to power the visitor counter, aligning with SDG 7's objective of ensuring access to affordable, reliable, sustainable, and modern energy for all.
2. **SDG 9: Industry, Innovation, and Infrastructure:** The project fosters innovation by utilizing piezoelectric technology in a practical application, contributing to advancements in infrastructure and technology development.
3. **SDG 11: Sustainable Cities and Communities:** The visitor counter provides valuable data for urban planners and policymakers to optimize infrastructure planning and resource allocation, enhancing the sustainability and efficiency of cities and communities.
4. **SDG 12: Responsible Consumption and Production:** By promoting the use of energy-efficient and renewable technologies, the project encourages responsible consumption and production practices, contributing to sustainable development efforts.